## 11.4.3 Distribution function (Cumulative distribution function)

**Definition 11.7 : (Cumulative Distribution Function)**

*The* ***distribution function or cumulative distribution function*** $F(x)$ *of a continuous random variable* $X$ *with probability density* $f(x)$ *is*

$$F(x) = P(X \le x) = \int_{-\infty}^{x} f(u)\,du, \qquad -\infty < u < \infty .$$

**Remark**

(1) In the discrete case, $f(a) = P(X = a)$ is the probability that $X$ takes the value $a$.

In the continuous case, $f(x)$ at $x = a$ is not the probability that $X$ takes the value $a$, that is $f(a) \neq P(X = a)$. If $X$ is continuous type, $P(X = a) = 0$ for $a \in \mathbb{R}$.

(2) When the random variable is continuous, the summation used in discrete is replaced by integration.

(3) For continuous random variable

$P(a < X < b) = P(a \le X < b) = P(a < X \le b) = P(a \le X \le b)$

(4) The distribution function of a continuous random variable is known as Continuous Distribution Function.

### 11.4.3.1 Properties of distribution function

For a continuous random variable X, the cumulative distribution function satisfies the following properties.

(i) $0 \le F(x) \le 1$.

(ii) $F(x)$ is a real valued non-decreasing. That is, if $x < y$, then $F(x) \le F(y)$.

(iii) $F(x)$ is continuous everywhere.

(iv) $\lim_{x \to -\infty} F(x) = F(-\infty) = 0$ and $\lim_{x \to \infty} F(x) = F(+\infty) = 1$.

(v) $P(X > x) = 1 - P(X \le x) = 1 - F(x)$.

(vi) $P(a < X < b) = F(b) - F(a)$.

**Example 11.11**

Find the constant $C$ such that the function $f(x) = \begin{cases} Cx^2 & 1 < x < 4 \\ 0 & \text{Otherwise} \end{cases}$

is a density function, and compute (i) $P(1.5 < X < 3.5)$ (ii) $P(X \le 2)$ (iii) $P(3 < X)$.

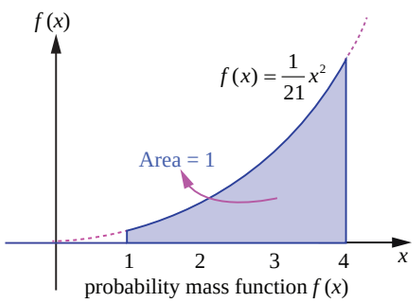
**Solution**

Since the given function is a probability density function,

$$\int_{-\infty}^{\infty} f(x)\,dx = 1.$$

That is $\int_{-\infty}^{1} f(x)\,dx + \int_{1}^{4} f(x)\,dx + \int_{4}^{\infty} f(x)\,dx = 1.$

From the given information,



probability mass function $f(x)$

Fig. 11.12



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